

Maru Metall — Unit Economics & Financial Model

SteelQuote AI · 28DIGITAL Business Challenge 2026

1. THE CORE ECONOMIC ARGUMENT

The business case for SteelQuote AI does not rest on speculative market sizing. It rests on a simple internal calculation: how many engineering hours are spent quoting projects, what is the cost of those hours, and what is the value of recovering them?

2. CURRENT STATE — COST OF MANUAL ESTIMATION

Volume and effort

Metric	Value	Source
Annual price inquiries received	500+	Maru challenge brief
Estimated effort per inquiry	Several hundred hours (total)	Maru challenge brief
Assumed average per inquiry	~8–20 hours	Engineering estimate: simple inquiry 4–8h; complex IFC+specs 20–40h
Senior estimator daily rate (est.)	€350–€500/day	Typical Nordic industrial rate
Total annual estimation cost (mid estimate)	500 x 12h x €55/h	~€330,000/year

The hidden cost

Not all 500 inquiries become orders. If Maru wins 30–40% of bids, then 300–350 inquiries per year are quoted and lost. Those are pure overhead — engineering hours spent generating no revenue. **AI assistance can dramatically reduce the cost of non-winning bids.**

3. REFERENCE PROJECT DATA

Example 4 — Pedestrian Steel Skybridge (HP11257-03)

Metric	Value
Project revenue	€591,285
Fabrication revenue	€576,885
Transport	€14,400

Metric	Value
Steel weight	206,414 kg
Paint system	C3.07 (all elements)

Element breakdown and unit prices

Element	Weight (kg)	Revenue (EUR)	Unit price (EUR/kg)
WI columns	72,143	222,073	3.08
WI beams	40,264	99,130	2.46
Beams	60,473	154,949	2.56
Bracings (horiz+vert)	22,389	59,523	2.66
Frame	4,357	12,820	2.94
CFJ parts (separately)	755	3,604	4.77
Erection/support plates	6,033	24,785	4.11
Total	206,414	576,885	avg 2.79

Key insight: Unit price varies 94% between cheapest (WI beams €2.46/kg) and most expensive (CFJ parts €4.77/kg). This variation reflects fabrication complexity — drilling density, weld length, surface-to-weight ratio — which is precisely the logic SteelQuote AI must encode.

Example 2 — Akkasater Storage Hall (TU4031)

Metric	Value
Project revenue	€333,395
Steel weight	148,315 kg
Phase 1+2 paint system	C2M
Phase 3	HDG (hot-dip galvanizing)

Element breakdown

Element	Weight (kg)	Revenue (EUR)	Unit price (EUR/kg)
Columns (C2M)	35,141	68,314	1.94
WQ beams (C2M)	11,381	22,637	1.99
Beams (C2M)	75,914	131,863	1.74
Bracings (C2M)	12,601	23,161	1.84
Window/door frames (C2M)	10,276	18,877	1.84
Outside entrance (HDG)	2,714	6,413	2.36
Indoor bollards (HDG)	288	954	3.31
Bracings D30 w. turnbuckles	72 pcs	17,564	243.94/pc
Hilti anchors	150 pcs	935	6.23/pc
Fasteners	1 set	5,302	—
Transport	1 set	37,375	—
Total	148,315 kg	333,395	avg 2.25

4. COST COMPONENT STRUCTURE

Based on the `Basic cost estimation principles.xlsx` provided by Maru, a correct cost estimate must cover:

4.1 Steel material

Component	Unit	Notes
Material purchasing cost	EUR/kg	Base steel price by grade (S235, S355, S275)
Material overconsumption	EUR/kg	Profiles: depends on stock length vs. element lengths; Plates: depends on detail shape vs. plate dimensions

4.2 Detail preparation (fabrication labour)

Operation	Unit	Notes
Plasma cutting of plates	min/m	Varies by plate thickness
Cleaning of plasma cut edges	min/m	Always after plasma cut
Rounding of sharp edges	min/m	Varies by corrosion class requirements
Hole drilling — plates	min/pcs	Depends on hole diameter and depth
Hole drilling — profiles	min/pcs	Depends on hole diameter and depth
Profile sawing	min/pcs	Cross-section dependent
Profile plasma cutting with robot	min/m	—
Profile bolt-hole plasma cutting	min/pcs	—

4.3 Welding

Operation	Unit	Notes
Assembling	min/pcs	Tack welding to shape; size and weight dependent
Welding	min/m	Converted to equivalent a5 fillet weld length; position and quality dependent
Weld splicing	min/m	Plate/profile splicing

4.4 Surface treatment

System	Unit	Notes
Shot-blasting	EUR/m2	Pre-treatment before painting
Paint systems C2M through C4H	EUR/m2	Different layer thickness and paint type per class
Fire protection R30/R60/R90	EUR/m2	Complex; profile/temperature dependent
Hot-dip galvanizing (HDG)	EUR/kg	ISO 1461

4.5 Transport and packaging

Item	Unit	Notes
Packaging	EUR/kg	Standard Maru packing procedure
Transport	EUR/truck	Standard (13.5m) vs. special (20m) trucks

5. AI ESTIMATION ACCURACY — TARGET METRICS

Benchmark: what “good” looks like

Based on the two reference projects:

Metric	Target
Total cost estimate accuracy vs. actual	Within $\pm 10\%$ on first pass
Element-level unit price accuracy	Within $\pm 15\%$ per element category
Labour hour estimate accuracy	Within $\pm 20\%$ (most variable component)
Surface treatment cost accuracy	Within $\pm 5\%$ (most formulaic component)
Assumption flagging completeness	100% — every gap must be documented

6. BUSINESS CASE FOR MARU

Conservative scenario: AI handles 30% of inquiries with 50% time reduction

Metric	Value
Inquiries per year	500
Inquiries AI assists (30%)	150
Average hours saved per inquiry (50% of 12h)	6h
Total hours recovered	900 h
Value at €55/h (blended estimator rate)	€49,500/year

Base scenario: AI handles 50% of inquiries with 60% time reduction

Metric	Value
Inquiries AI assists	250
Hours saved per inquiry	7.2h
Total hours recovered	1,800 h
Value at €55/h	€99,000/year

Optimistic scenario: AI handles 70% with 70% time reduction

Metric	Value
Inquiries AI assists	350
Hours saved per inquiry	8.4h
Total hours recovered	2,940 h
Value at €55/h	€161,700/year

PoC build cost (minimal viable version)

Cost item	Estimate
LLM API and IFC parsing (3 months)	€3,000–€6,000
Developer time (PoC build)	€8,000–€15,000
Estimator validation time	€2,000–€4,000
Total PoC investment	€13,000–€25,000
Payback period (base scenario)	2–3 months

The PoC investment is recovered in less than one quarter of operation at the base scenario efficiency level.

7. KEY NUMBER FOR THE PITCH

500 inquiries × 12 hours = 6,000 engineering hours per year. SteelQuote AI recovers up to 60% of those hours. Payback in 2–3 months. Every hour saved above that is pure capacity.